

Farjana Kabir

LLM Engineer | Generative AI & NLP Specialist

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Summary

LLM Engineer with 2+ years of experience designing, fine-tuning, and deploying NLP and generative AI solutions. Proficient in Retrieval-Augmented Generation (RAG), agentic workflows, embeddings, prompt engineering, and scalable NLP pipelines. Skilled in transformer-based architectures, Hugging Face Transformers, and LangChain. Hands-on experience in developing domain-specific LLMs and multimodal AI applications from research to production. Familiar with cloud platforms (AWS) and containerized deployments.

Core Skills

LLM & NLP: Large Language Models (GPT, Gemini, LLaMA), RAG, Agentic Workflows, Prompt Engineering, Fine-tuning (LoRA, PEFT), Embeddings, Tokenization, Hugging Face Transformers, LangChain

Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Keras

Data Processing & Pipelines: Python, Pandas, NumPy, Vector Databases (FAISS, ChromaDB, AstraDB), ETL

Cloud & Deployment: AWS, Docker, FastAPI, Git, CI/CD, Containerization Concepts

Evaluation & Metrics: BLEU, ROUGE, F1 Score, Accuracy

Traditional NLP: N-gram models, TF-IDF, Stemming/Lemmatization, Rule-based Processing

DevOps & Systems: Version Control, Logging, Monitoring, Microservices Architecture

Professional Experience

Senior Data Scientist

somikoron.ai

Sep 2023 – Feb 2025

- Designed and deployed Bengali Question-Answer Generation System using RAG architecture, custom embeddings, and open-source LLMs, improving question accuracy by 15%.
- Developed AI-powered Prescription Parser with multimodal LLMs and domain-specific prompt engineering, achieving 70% extraction accuracy.
- Built and fine-tuned an Object Detection System (YOLO-based) for government use, ensuring scalable and high-precision deployments.
- Implemented Bengali rule-based stemmer reducing unique vocabulary count by 3.34%, improving downstream NLP model performance.
- Designed optimized data structures for ML-based recommender systems, boosting performance by 10%.

Data Scientist

somikoron.ai

Mar 2023 – Aug 2023

- Built custom NLP applications for Bengali language processing.
- Optimized large-scale data cleaning pipelines for training datasets.
- Extracted ~50 data points from unstructured sources using web scraping and NLP parsing.

Projects

Question-Answering System with Agentic RAG: Built modular RAG pipeline with agentic tool orchestration, improving domain Q&A accuracy. Applied role-based guardrails to reduce hallucinations.

Ecommerce Product Assistant: Designed assistant using embeddings + vector search for semantic queries; deployed via scalable inference API.

QuickSum: Developed summarization pipeline with fine-tuned transformers; benchmarked with ROUGE and optimized for lightweight deployment.

Bilingual Health QnA: English–Bengali Q&A system with fine-tuned transformers and low-rank adaptation (LoRA).

Multi-purpose Personal Assistant: Multimodal assistant integrating Gemini API, FastAPI, and LangChain workflows. Reduced app creation time by 85%. Applied LLM tool calling and orchestration for multimodal inputs.

Education

B.Sc. in Computer Science & Engineering Khulna University of Engineering & Technology (KUET)

2017 – 2022

Concentration: Machine Learning

Relevant Coursework: NLP, Data Mining, Machine Learning, Image Processing, Biomedical Engineering